38.

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE, Final Examination, Autumn 2024
Course Code: CSE 2427, Course Title: Theory of Computing

Total marks: 50 *Time: 2 hour 30 minutes*

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

Group A

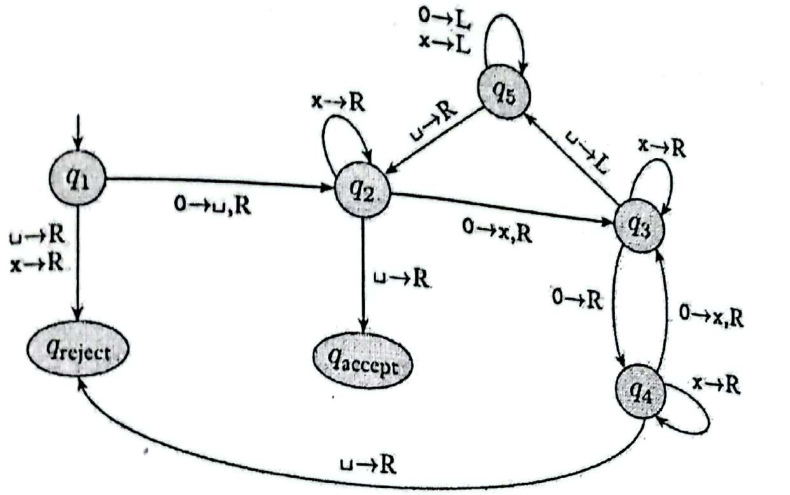
1.a)	Construct a Context-Free Grammar (CFG) that generates the following language over alphabet Σ {0,1}- $L = \{w \mid w = w^R, that\ is, the\ string\ w\ is\ a\ palindrome\}$ *Or,* Construct a Context-Free Grammar (CFG) equivalent to the regular expression over alphabet Σ {0,1} representing the language $(0+1)^*011^*$.	3	CO1	Ap
b)	Give the formal definition of Context-Free Grammar (CFG) and Chomsky Normal Form.	2	CO2	R
c)	Consider the following grammar – $S \to E,\ E \to E + E \mid E - E \mid E * E,\ E \to (E) \mid a$ Show that this grammar is ambiguous. In particular show that the string "a+a*(a-a)" has two i)parse trees and ii)Leftmost derivations or Rightmost derivations.	5	CO2	An
2.a)	Using the pumping lemma show that the following languages are not context free. $L1 = \{a^n b^n c^i \mid i \leq n\ and\ i, n \geq 0\}$ and $L2 = \{x^n y^n z^n \mid n \geq 1\}$	3	CO1	U
b)	How can Context-Free Grammar (CFG) be simplified? Write down the procedure for eliminating unit productions from a CFG. Remove the unit productions from the following grammar : S→AC, A→a , C→B | d, D→E, E→b *Or,* Can you give a context-free grammar (CFG) for the language, $L = \{a^{2n} b^{2n} c^{4n} \mid n \geq 0\}$ over the alphabet $\Sigma = \{a, b\}$, If you cannot, justify the reason.	3	CO2	Ap
c)	Convert the following CFG into an equivalent CFG in Chomsky Normal Form: $S \to ASA \mid aB, \quad A \to B \mid S, \quad B \to b \mid \epsilon$ *Or,* Convert the following CFG into an equivalent CFG in Chomsky Normal Form: $S \to ASB, \quad A \to aAS \mid a \mid \varepsilon, \quad B \to SbS \mid A \mid bb$	4	CO2	Ap

Group B

3.a)	What is a Pushdown Automaton (PDA)? Explain the role of the stack in a PDA. Why is it necessary for some languages? *Or,* Write down the difference between Finite Automata and Push Down Automata.	2	C01	U
b)	Construct PDAs that recognize the following languages: $L1 = \{a^i b^j c^k \mid i, j, k \geq 0\ and\ i = j\ or\ j = k\}$ L2 = $\{ww^R$ | w ε {a,b}*}, where W^R is the reverse of w.	5	CO3	Ap

4. Consider the following Turing machine.

a)



4 CO3 A

Give the sequence of configurations that the machine enters when started with the following strings.

i. 0000

ii. 000000

b) Give the implementation-level description of Turing Machine that accepts the following languages (*any two*): 6 CO3 C

(i) **L = {w | w is the set of strings with an equal number of 0's and 1's}.**

(ii) **L = {ww^R | w is any string of 0's and 1's}.**

(iii) **L = n − 1, where n > 0.**

5.

a) Convert any one of the following CFG into an equivalent CFG in Chomsky normal form. 4 CO2 A

S → aSb | bY | Ya
Y → bY | aY | c | ε

OR

S → aXbX
X → aY | bY | ε
Y → X | c

b) Find out whether the following problem is decidable or not. 3 CO4 E

Is a number 'm' prime?

OR

Determining if a given graph G is connected.

c) What are NP-complete and NP-Hard problems? How can you show that a problem is NP-complete? 3 CO5 U

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Term Examination, Spring 2024

Course Code: CSE 2425 **Course Title: Theory of Computing**

Total marks: 50 *Time: 2 hour 30 minutes*

[Answer all the questions. Figures in the right-hand margin indicate full marks. The questions must be answered in order.]

Group A

1.a)	Construct a **Context-Free Grammar (CFG)** that generates the following language over the alphabet $\Sigma = \{0,1\}$ $L = \{w \mid \text{the length of } w \text{ is odd and its middle symbol is a '0'}\}$ *Or,* Construct a **Context-Free Grammar (CFG)** corresponding to the regular expression $(0+1)^*011^*$ over the alphabet $\Sigma = \{0,1\}$. That is, any string described as ***"any combination of '0' and '1' followed by '01' ending with any number of '1's"*** belongs to the associated language.	3	CO1	Ap
1.b)	Give the formal definition of **Context-Free Grammar (CFG)** and **Chomsky Normal Form (CNF).**	2	CO2	R
1.c)	When is a grammar called *ambiguous*? Consider the following grammar $S \rightarrow SAB \mid \epsilon, \quad A \rightarrow AaB \mid a, \quad B \rightarrow AS \mid b$ This grammar is **ambiguous.** Show in particular that the string **aaaabaabbabab** has two: i. Parse trees ii. Leftmost derivations iii. Rightmost derivations	5	CO2	An
2.a)	Using the **pumping lemma** show that the following languages are not context free (any one): $L = \{a^n b^n c^i \mid i \leq n, i \geq 0, n \geq 0\}$, $L = \{x^n y^n z^n \mid n \geq 1\}$ *or,* Can you give a context-free grammar (CFG) for the following language over the alphabet $\Sigma = \{a, b\}$ – **All strings in the language** $\{a^n b^{2n} c^{4n} \mid n \geq 0\}$ If you cannot, justify the reason.	3	CO1	U
2.b)	How can Context-Free Grammar (CFG) be simplified? Write down the procedure for **eliminating unit productions** from a CFG. **Remove the unit productions** from the following grammar – $S \rightarrow AC, \quad A \rightarrow a, \quad C \rightarrow B \mid d \quad D \rightarrow E, \quad E \rightarrow b$	3	CO2	Ap
2.c)	Convert the following CFG into an equivalent CFG in **CNF:** $S \rightarrow TX$ $T \rightarrow 0T0 \mid 1T1 \mid \#X$ $X \rightarrow 0X \mid 1X \mid \epsilon$ *Or,* Convert the following CFG into an equivalent **CNF** form. S → ASB A → aAS|a | ε B → SbS |A|bb	4	CO2	Ap

Group B

3.a)	Formally define the pushdown automata. Illustrate the component of PDA with figure. *Or,* Why do you think pushdown automata are more powerful than finite automata?	2	C02	R
3.b)	Construct a **pushdown automaton** that recognizes the following language $L = \{t^i s^j a^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } j = k\}$ *Or,* Suppose the PDA $P = (\{q, p\}, \{0, 1\}, \{Z_0, X\}, \delta, q, Z_0, \{p\})$ has the following transition functions: $\delta(q, 0, Z_0) = \{(q, XZ_0)\}$, $\delta(q, 0, X) = \{(q, XX)\}$, $\delta(q, 1, X) = \{(q, X)\}$, $\delta(q, \epsilon, X) = \{(p, \epsilon)\}$, $\delta(p, \epsilon, X) = \{(p, \epsilon)\}$, $\delta(p, 1, X) = \{(p, XX)\}$, $\delta(p, 1, Z_0) = \{(p, \epsilon)\}$ Starting from the initial instantaneous description (q, w, Z_0), show all the reachable instantaneous descriptions when the input w is: i. 0011 ii. 010	4	CO3	An

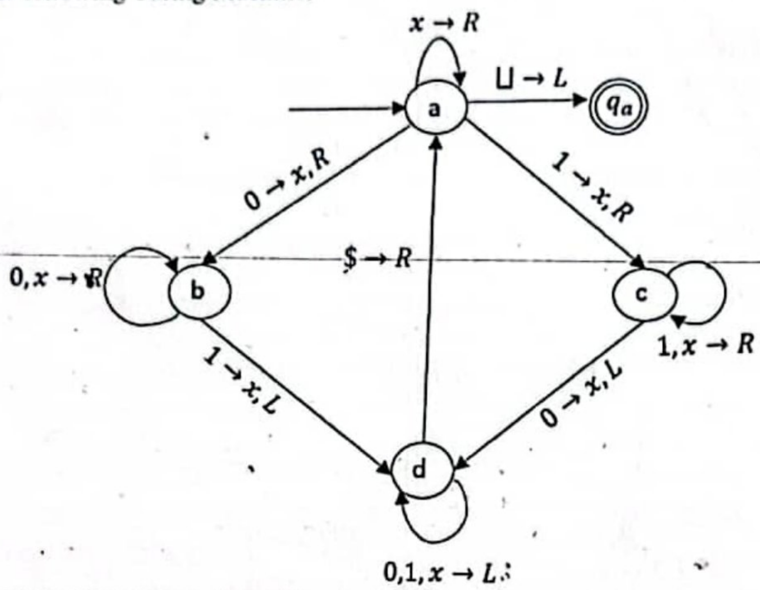
3.c) Convert any one of the following CFGs to an equivalent pushdown automaton 4 CO3 Ap

$$S \rightarrow TT|U$$
$$T \rightarrow 0T|T0|\#$$
$$U \rightarrow 0U00|\#$$

Or,

$$S \rightarrow aXbY$$
$$X \rightarrow aYa \mid \epsilon$$
$$Y \rightarrow bXb \mid c \mid \epsilon.$$

4.a. Consider the following Turing Machine. 4 CO3 U



Give the **sequence of configurations** that the machine enters when started with the following strings.

i. 0100

ii. 011010

4.b. Give the **implementation-level description** of Turing Machine that decides the following languages (*any two*) 2 * 3 = 6 CO3 An

i. L = {w | w does not contain twice as many 0s as 1s}

ii. L = $\{ww^R \mid$ w is any string of 0's and 1's$\}$

iii. L = $\{a^n b^m c^n d^m | n \geq 1, m \geq 1\}$

5.a. What is the Church-Turing Thesis?
Define *decidability* and *undecidability* in the context of *theory of computing*. 4 CO4 Ap

5.b. Prove that the **Halting Problem** for Turing Machine is undecidable. 3 CO5 Ap

Or,

Consider the language A_{DFA} = {(B , w) | B is a DFA that accepts input string w}.
Prove that, "*A_{DFA} is a decidable language*".

5.c. Define the complexity classes P, NP, NP-hard and NP-complete. How can you show that a problem is NP-complete? 3 CO6 U

OR

What is **reducibility**? What do you understand by **polynomial time complexity** and **exponential time complexity**? Explain with an example.

CS CamScanner

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
Dept. of Computer Science & Engineering
B.Sc. in CSE Semester Final Examination, Autumn 2023
Course Code: CSE-2425 Course Title: **Theory of Computing**
Full Marks: 50 Time: **2 Hours 30 minutes**
[Answer the questions from Group-A and Group-B, using separate answer script.]

Group – A [20 Marks]

1. a) Define context-free grammar (CFG). Consider the following CFG: 1 CO3 U

$$G_1: \quad S \rightarrow aABe$$
$$A \rightarrow Abc|b$$
$$B \rightarrow d$$

i) Identify the elements/components of G_1, when formally denoted as a 4-tuple. 1 U
ii) Give the leftmost derivation for the string: "$abbcbcde$". 1 CO3 C
iii) Construct the parse tree with G_1 for the input string: "$abbcde$" 1 Ap

Or

What do you mean by ambiguity? Consider the following CFG: 1 U

$$G_2: \quad S \rightarrow E$$
$$E \rightarrow E + E|E - E|E * E$$
$$E \rightarrow (E)|a$$

i) Is this grammar G_2 ambiguous? If yes, prove that using an appropriate string belonging to the language $L(G_2)$. 1 U
ii) Describe the language $L(G_2)$ in English. 1 U
iii) Give the rightmost derivation for the input string: "$a + a * (a - a)$" 1 C

b) Find the CFGs that generate the following languages over the alphabet $\Sigma = \{0,1\}$ 1+1 CO3 U

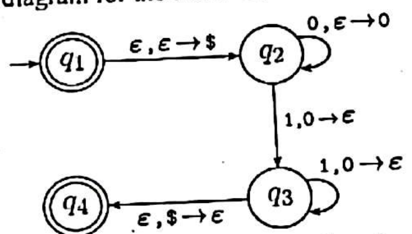
i) $L_1 = \{w|$ length of w is odd and its middle symbol is '1'$\}$
ii) $L_2 = \{w|\ w = w^R$, that is, the string w is a palindrome.$\}$

c) What is Chomsky normal form (CNF)? Transform the following CFG into an equivalent CFG in CNF: 1+3 CO3 An

$$G_3: S \rightarrow ASA|aB$$
$$A \rightarrow B|S$$
$$B \rightarrow b|\varepsilon$$

2. a) Explain the general notion of pushdown automata (PDA) with the schematic diagram and present its formal definition as a tuple. 2 CO3 R

b) Consider the following state diagram for the PDA P_1: 4



An
U
An

i) Identify the elements of this PDA P_1, denoted as a 6-tuple.
ii) Give a concise description in English of the language L that is recognized by the above PDA P_1. Is this language L_3 regular? CO3 C
iii) Is the PDA P_1 deterministic or nondeterministic? Why? An

Or

Construct a PDA that recognizes the following language

$$L = \{a^i b^j c^k \mid i,j,k \geq 0 \ and\ i + j = k\}$$

Give an informal description of the PDA, and illustrate how it proceeds with the input string "abbccc".

c) State the **pumping lemma** for context-free languages. Using this version of the **pumping lemma** show which of the following language(s) is (are) not context free. 3 CO2 Ap

$$L = \{a^n b^m c^n d^m \mid n, m \geq 0\}$$
$$L = \{ww \mid w \in \{0,1\}^*\}$$

d) For the following CFG, show the state diagram of an equivalent PDA.

$E \rightarrow E + T \mid T$
$T \rightarrow T * F \mid F$
$F \rightarrow a$

Or

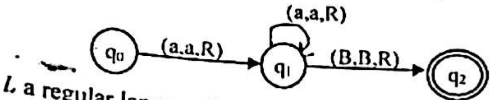
Remove ε-production from the following CFG:

$S \rightarrow aB \mid bA$
$A \rightarrow Bb \mid a$
$B \rightarrow b \mid \varepsilon$

Group – B [30 Marks]

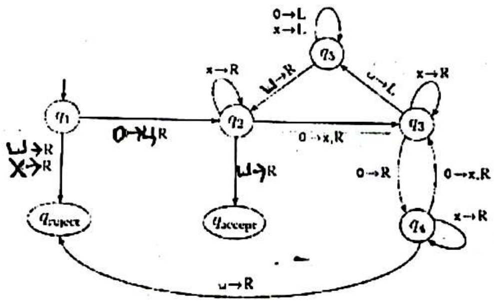
3. a) Explain the notion of the Turing machine (TM) with the schematic diagram and introduce the elements of the tuple that formally defines it. Differentiate between PDA and TM. 3 CO3 R, U

b) Give the language L for the TM represented by the following state diagram: 2 CO3 Ap



Is L a regular language?

c) Consider the following Turing Machine. 5 CO3 An



CO3 An

Give the *sequence of configurations* that the machine enters when started with the following strings. i. 00 ii. 000000

Or

Give the *implementation-level description* of Turing Machines that decide the following languages (*any two*)

i. $L = \{w \mid w$ is a string with 0s and 1s and contains 1s in a multiple of 3 $\}$

ii. $L = \{ww^R \mid w$ is any string of 0's and 1's $\}$

iii. $L = \{a^n b^n a^n \mid n \geq 0\}$

4. a) Let the language A_{DFA} be described as $A_{DFA} = \{\langle B, w\rangle \mid B$ is a DFA that accepts input string $w\}$. Prove that A_{DFA} *is a decidable language.* 4 CO4 C

Or

Let the acceptance language of NFA $A_{NFA} = \{\langle N, w\rangle \mid N$ is an NFA that accepts input string $w\}$. Show that A_{NFA} is decidable.

b) Introduce the *diagonalization method/argument*. Produce an illustrative example/application of it to show that the set of real numbers is uncountable. 3 CO4 An

c) What is decidability? Let a language L is defined as $L = \{\langle M, w\rangle \mid M$ *is a TM and M halts on input w*$\}$. Show that L is undecidable. 3 CO4 Ap

5. a) What are the key ideas to formally prove that *every non-deterministic TM has an equivalent deterministic TM*? 4 CO4 Ap

b) What is polynomial-time reducibility? 1 CO5 R

d) Formally define the classes P, NP, NP-Hard and NP-complete. How can you show that a problem is NP-complete? 5 CO6 U

[END]

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2022

Course Code: CSE-2425 Theory of Computing

Total marks: 50 *Time: 2 hours 30 minutes*

[Figures in the right-hand margin indicate full marks

Course Outcomes and Bloom's taxonomy levels are mentioned in additional columns]

Bloom's Taxonomy Levels (Cognitive Domain)						
Letter Symbols	R	U	A	N	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

CO DL

1.

a) Describe the components of context-free grammar. Is any of the components have a similarity with any of the components of regular language or finite automata? **CO1 U 3**

OR

Define and differentiate between the following.

a. Derivation and Parse Tree
b. Leftmost and Rightmost derivation

b) What is ambiguity? Determine whether the following grammar is ambiguous. **CO2 N 3**

$S \to AB$

$A \to aA \mid abA \mid \varepsilon$

$B \to bB \mid abB \mid \varepsilon$

c) Consider the following grammar **CO2 A 4**

$S \to SSx \mid SSy \mid SSz \mid a \mid b \mid c$

Show how to derive the string **cabxycz** using this grammar using a left-most derivation. Draw the parse tree for the string.

OR

Show how to derive the string **cabxycz** using this grammar using a right-most derivation. Draw the parse tree for the string.

2.

a) Show using the pumping lemma which of the following languages are context-free. **CO2 N 3**

i. $L1 = \{w \mid w \in a^n b^n c^{2n} \mid n \geq 0\}$
ii. $L2 = \{w \mid w \in a^n b^n c^n \mid n \geq 0\}$

OR

Describe the following languages using a context-free grammar.

a. 0*1*
b. 1(01)*
c. (11 U 0)*

b) Give a context-free grammar (CFG) for each of the following languages over the alphabet **Σ = {a,b}:-** **CO2 C 4**

i. $L = \{a^{2n} b^m c^n \mid n \geq 0\}$
ii. $L = \{a^m b^n c^{m+n} \mid n \geq 0\}$

c) Convert the following CFG into an equivalent CFG in Chomsky normal form. **CO2 N 3**

$R \to aSa \mid bRb \mid S$

$S \to aTb \mid bTa \mid aS$

$T \to XTX \mid X \mid \varepsilon$

$X \to a \mid b$

Group-B

3.

a) How ε-rules are removed when converting a grammar to Chomsky normal form? **CO2 U 2**

OR

Why do you think pushdown automata are more powerful than finite automata?

b) Construct a pushdown automaton that recognizes the following language **CO3 C 4**

$L = \{a^{2n}b^{m}c^{n} \mid n \geq 0\}$

OR

Construct a pushdown automaton that recognizes any arithmetic expression involving +, *, and any one-digit integer.

c) Convert any one of the following context-free grammar (CFG) to an equivalent pushdown automaton **CO3 N 4**

$S \rightarrow XYm \mid XYn$
$X \rightarrow aX \mid \varepsilon$
$Y \rightarrow bY \mid \varepsilon$

OR

$S \rightarrow aXbY$
$X \rightarrow aYa \mid \varepsilon$
$Y \rightarrow bXb \mid c \mid \varepsilon$

4.

a) *What is the Church-Turing thesis?* **CO4 U 2**

b) *Can you run a nondeterministic algorithm on a deterministic machine instead of a nondeterministic one? If your answer is yes, then explain how you can do it and how the running time will be affected. If your answer is no, then explain why it will not be possible.* **CO5 E 4**

c) *Give the implementation-level description of a Turing machine that decides the following languages:* $L = \{a^{2n}b^{n}c^{n} \mid n \geq 0\}$. **CO3 A 4**

5.

a) Let the language $A_{DFA} = \{(B, w) \mid B$ is a DFA that accepts input string $w\}$. Prove that "A_{DFA} is a decidable language." **CO4 E 3**

b) Show that the set of infinitely long binary sequences is uncountable. **CO4 N 3**

c) What are N Let, **CO5 N 4**

$A_{TM} = \{\langle M,w\rangle \mid M$ is a TM and M accepts $w\}$

and

$HALT_{TM} = \{\langle M,w\rangle \mid M$ is a TM and M halts on input $w\}$

Prove that if A_{TM} is undecidable then $HALT_{TM}$ is also undecidable.

CS CamScanner

Shahrian Afti

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2022

Course Code: CSE-2425 Theory of Computing

Total marks: 50 *Time: 2 hours 30 minutes*

[Figures in the right-hand margin indicate full marks
Course Outcomes and Bloom's taxonomy levels are mentioned in additional columns]

Bloom's Taxonomy Levels (Cognitive Domain)						
Letter Symbols	R	U	A	N	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

			CO	DL
1.				
a)	Construct a Context-Free Grammar (CFG) for the regular expression **(0 + 1)*0 1***, that is, any combination of **0** and **1** followed by a single **0** and ending with any number of **1**. **OR,** Construct a CFG for the regular expression **0*1(0 + 1)***, that is, any number of **0** followed by a single **1** and ending with any combination of **0** and **1**.	3	CO1	A
b)	Consider G whose productions are **S → aAS/a,** **A → SbA/SS/ba.** Show that **S → aabbaa** by constructing a derivation tree, by rightmost derivation, whose yield is **aabbaa.**	2	CO2	A
c)	Consider the grammar **S → iCtS/iCtSeS/a** **C → b** This grammar is ambiguous. Show in particular that the string **ibtibtaea** has two: (i) parse trees, (ii) leftmost derivations, (iii) rightmost derivations.	5	CO2	N
2.				
a)	How do you use the pumping lemma to determine if a language is context-free? **OR** Can you describe a regular language using context-free grammar? Illustrate using an example.	3	CO1	N

b)	Can you give a context-free grammar (CFG) for the following languages over the alphabet $\Sigma = \{a,b\}$:- All strings in the language L: $\{a^n b^{2n} c^{4n} \mid n \geq 0\}$ If you cannot, justify the reason.	3	CO2	E
c)	Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{a,b\}$:- - All strings in the language L: $\{a^n b^{2n} \mid n \geq 0\}$ - All nonempty strings that read the same from left or right.	4	CO2	C

OR

How can Context-Free Grammar (CFG) be simplified? Write down the procedure for eliminating unit productions from a CFG. Remove the unit productions from the following grammar:
$S \rightarrow AB$, $A \rightarrow a$, $B \rightarrow C/b$, $C \rightarrow D$, $D \rightarrow E$, $E \rightarrow a$.

Group-B

3.

a)	What are ε-rule and unit rules?	2	CO2	U
b)	Construct a pushdown automaton that recognizes the following language L: $\{a^n b^{2n} \mid n \geq 0\}$	4	CO3	C/E

OR

Suppose the PDA $P = (\{p, q, f\}, \{0, 1\}, \{0, 1, Z_0\}, \delta, q, Z_0, \{f\})$ has the following transition functions:

$\delta(q, 0, Z_0) = \{(q, XZ_0)\}$, $\delta(q, 0, X) = \{(q, XX)\}$,
$\delta(q, 1, X) = \{(p, \varepsilon)\}$, $\delta(p, 1, X) = \{(p, \varepsilon)\}$,
$\delta(p, \varepsilon, Z_0) = \{(f, Z_0)\}$.

Starting from the initial ID (q, w, Z_0), show all the reachable ID's when the input w is:
(i) 00001111
(ii) 00011

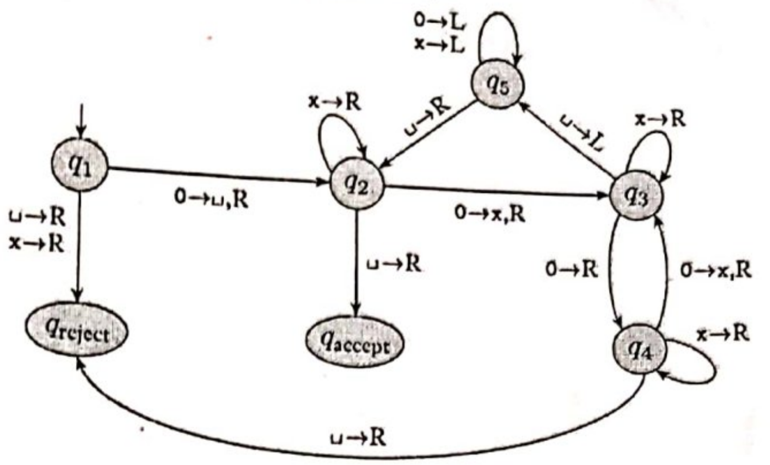
c)	Convert any one of the following context-free grammar (CFG) to an equivalent pushdown automaton	4	CO1	N

$S \rightarrow aSb \mid bY \mid Ya$
$Y \rightarrow bY \mid aY \mid c \mid \varepsilon$

OR

$S \rightarrow aXbX$
$X \rightarrow aY \mid bY \mid \varepsilon$
$Y \rightarrow X \mid c$

4. a) Consider the following Turing machine. 4 CO3 A



Give the sequence of configurations that the machine enters when started with the following strings.

i. 0000
ii. 000000

b) Give the implementation-level description of Turing Machine that accepts the following languages (*any two*): 6 CO3 C

(i) **L = {w | w is the set of strings with an equal number of 0's and 1's}.**
(ii) **L = {ww^R | w is any string of 0's and 1's}.**
(iii) **L = n − 1, where n > 0.**

5. a) Convert any one of the following CFG into an equivalent CFG in Chomsky normal form. 4 CO2 A

$S \rightarrow aSb \mid bY \mid Ya$
$Y \rightarrow bY \mid aY \mid c \mid \varepsilon$

OR

$S \rightarrow aXbX$
$X \rightarrow aY \mid bY \mid \varepsilon$
$Y \rightarrow X \mid c$

b) Find out whether the following problem is decidable or not. 3 CO4 E

Is a number 'm' prime?

OR

Determining if a given graph G is connected.

c) What are NP-complete and NP-Hard problems? How can you show that a problem is NP-complete? 3 CO5 U

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2021

Course Code: CSE-2425 Theory of Computing

Total marks: 50 Time: 2 hours 30 minutes

[Figures in the right hand margin indicate full marks]

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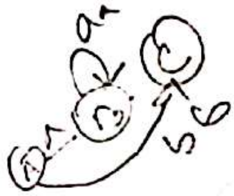
Bloom's Taxonomy Levels (Cognitive Domain)						
Letter Symbols	R	U	A	N	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

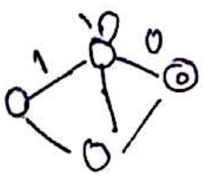
1.					
a)	Convert the following regular expressions to NFA. i) a(aa)* U (bb)* ii) a(a U b)*a U b(a U b)*b	2	CO1	A	
b)	What is ambiguity? Determine whether the following grammar is ambiguous. $S \rightarrow AB$ $A \rightarrow aA$ $A \rightarrow abA$ $A \rightarrow \varepsilon$ $B \rightarrow bB$ $B \rightarrow abB$ $B \rightarrow \varepsilon$	3	CO2	N	
c)	Show how to derive the string **aabab** using this CFG using a left-most derivation. Draw the parse tree for the string. **Or** Show how to derive the string **abaabb** using this CFG using a right-most derivation. Draw the parse tree for the string.	5	CO2	A	
2.					
a)	Differentiate between DFA and NFA. Construct a DFA that recognize the following language. In all parts , the alphabet is {0,1}: *{w | w begins with a 1 and ends with 0 }* **Or** Construct a Deterministic Finite Automata , $\sum$ = {a,b} and L(M)= {ω| ω Starts and ends with different symbol}.	5	CO1	A	
b)	Define simplification of CFG with types. Write down the procedure for removal of unit production with example.	5	CO2	R	

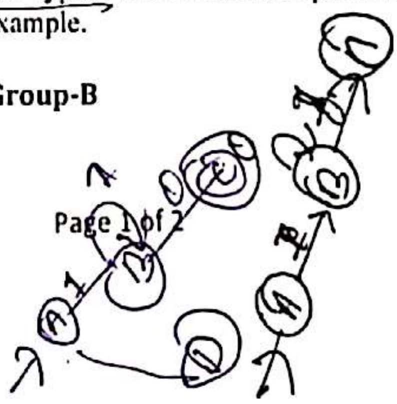


AB
aAB
aεB
aabB
aab

S → AB
→ aAB
→ aaAB
→ aaB
→ aabB
→ aab

Group-B





3.

a) i) Using pumping lemma for context free languages show that the following language is not context free. — 2+4 | CO2 | A

$L = \{w \mid w \in a^n b^n b^n\}$

ii) Convert any one of the following CFG into an equivalent CFG in Chomsky normal form.

$R \rightarrow aRb \mid bRb \mid S$

$S \rightarrow aTa \mid bTa$

$T \rightarrow XTX \mid X \mid \varepsilon$

$X \rightarrow a \mid b$

Or

Prove the theorem: A language is context free iff some pushdown automata recognize it. — 6 | CO2 | E

b) Remove null production from the following grammar: — 4 | CO1 | N

S---->ABAC , A--->aA|Є , B->bB|Є , C--->c

Or

Convert any one of the following context-free grammar (CFG) to an equivalent pushdown automaton

$R \rightarrow XRX \mid S$

$S \rightarrow aTa \mid bTb$

$T \rightarrow bTa \mid abTb \mid X \mid \varepsilon$

$X \rightarrow a \mid b$

4.

a) Give the implementation-level description of Turing machine that decides the following languages (any two) — 6 | CO3 | C

i. $\{w \mid w \text{ contains three times as many 1s as 0s}\}$

ii. $B = \{0^m 1^n 2^{m-n} \mid m, n > 0 \text{ and } m > n\}$

iii. $\{w \mid w \text{ is a string with 0s and 1s and contains 1s in a multiple of 3}\}$

iv. $D = \{0^i 1^j 2^k \mid i+k = 2 * j \text{ and } i,j,k > 0\}$

b) Define decidable, language. Find out whether the following problem is decidable or not: Is a number 'm' prime? — 4 | CO1 | A

5.

a) Differentiate between a finite automaton and a Turing machine — 2 | CO4 | U

b) Define the classes P, NP and NP-complete. Why NP-complete class is significant regarding the question whether P = NP? — 3 | CO5 | U

c) Show the relationship among the following types of language in a diagram: Regular language, context free language, decidable language. — 2 | CO4 | U

d) Can you run a nondeterministic algorithm on a deterministic machine instead of a nondeterministic one? If your answer is yes, then explain how you can do it and how the running time will be affected. If your answer is no, then explain why it will not be possible. — 3 | CO4 | E